

GWANGJU AP, Korea, Republic of

WMO: 471580

Lat: 35.126N

Lon: 126.809E

Elev: 12

StdP: 101.18

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.1	(c) -5.8	(d) -11.2	(e) 1.4	(f) -2.3	(g) -9.8	(h) 1.6	(i) -1.4	(j) 9.0	(k) 1.1	(l) 7.8	(m) 1.6	(n) 1.2	(o) 30	(p) 0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.4	(c) 34.1	(d) 26.3	(e) 32.9	(f) 25.6	(g) 31.2	(h) 24.8	(i) 27.1	(j) 31.6	(k) 26.5	(l) 30.9	(m) 26.0	(n) 30.1	(o) 3.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.1	25.2	20.3	28.4	24.9	19.9	28.2	85.5	31.6	82.7	30.9	80.4	30.3	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.5	(c) 5.7	(e) -11.2	(f) 35.3	(g) 1.8	(h) 1.2	(i) -12.6	(j) 36.1	(k) -13.6	(l) 36.8	(m) -14.7	(n) 37.4	(o) -16.0	(p) 38.3		
(a) 7.5	(b) 6.5	(c) 5.7	(e) -11.1	(f) 28.8	(g) 1.8	(h) 0.9	(i) -12.4	(j) 29.5	(k) -13.5	(l) 30.0	(m) -14.5	(n) 30.6	(o) -15.8	(p) 31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	1.2	2.5	6.8	13.0	18.0	22.9	26.9	27.3	22.1	15.4	8.8	3.6
	DBStd	9.57	3.11	3.68	3.25	3.70	2.56	2.00	2.55	2.03	3.02	3.29	3.52	2.93
	HDD10.0	867	273	211	106	13	0	0	0	0	0	3	62	200
	HDD18.3	2382	531	443	357	164	38	1	0	0	4	100	285	458
	CDD10.0	2366	0	1	7	103	247	387	523	537	364	170	27	1
	CDD18.3	838	0	0	0	4	27	137	264	278	118	9	0	0
	CDH23.3	7447	0	0	0	54	254	878	2536	2821	820	83	0	0
	CDH26.7	2633	0	0	0	3	39	248	981	1152	206	4	0	0
Wind	WSAvg	2.0	2.1	2.3	2.4	2.3	2.2	2.0	2.3	2.0	1.7	1.6	1.8	1.8
Precipitation	PrecAvg	1399	32	45	65	91	100	164	294	314	148	54	55	37
	PrecMax	1944	85	132	136	232	216	367	629	633	507	207	147	75
	PrecMin	827	6	11	16	18	25	36	53	83	24	3	4	3
	PrecStd	278	17	29	32	51	42	88	126	137	114	44	37	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	15.2	19.1	27.0	29.5	32.2	37.8	36.0	32.2	27.1	21.1	15.2
		MCWB	8.2	10.9	11.7	16.5	19.4	22.6	25.8	26.7	24.2	18.9	15.6	10.4
	2%	DB	9.9	12.8	17.0	25.0	27.8	30.9	34.3	34.2	30.2	25.8	19.2	13.0
		MCWB	7.5	8.6	10.5	16.4	18.3	22.0	26.4	26.3	23.2	18.0	13.6	8.7
	5%	DB	8.1	10.9	15.1	22.9	26.1	29.2	32.9	33.1	29.0	23.9	17.8	11.0
		MCWB	5.4	7.6	9.3	15.2	17.6	21.2	25.8	25.9	22.3	17.4	12.7	7.1
	10%	DB	6.2	9.0	13.2	20.8	24.8	28.0	31.2	31.9	27.7	22.0	15.8	9.2
		MCWB	3.8	5.8	8.3	14.2	17.1	21.0	25.4	25.3	21.8	16.2	12.0	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.8	14.7	20.6	21.7	25.4	27.7	28.3	25.8	21.1	17.3	11.4
		MCDB	10.5	14.1	16.2	23.6	25.1	28.5	32.4	31.5	28.7	23.8	19.3	14.5
	2%	WB	7.9	9.9	11.7	18.4	20.3	24.3	26.9	27.3	24.7	19.6	15.2	9.7
		MCDB	9.2	11.5	15.6	22.0	24.8	26.8	31.8	31.4	28.6	22.9	18.0	11.7
	5%	WB	5.7	7.7	10.3	16.3	19.1	23.5	26.4	26.7	23.9	18.6	13.7	7.9
		MCDB	7.4	10.2	13.5	21.1	23.6	26.3	31.0	31.0	27.4	22.2	16.2	10.4
	10%	WB	4.3	5.9	9.0	14.8	18.1	22.4	25.9	26.2	22.9	17.3	12.3	6.4
		MCDB	6.2	8.5	12.8	19.6	22.6	26.0	30.2	30.4	26.6	21.1	15.4	9.0
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.4	11.2	13.0	11.9	9.1	7.4	8.4	9.9	12.5	11.6	10.1
		MCDBR	10.2	11.9	14.6	16.0	15.2	12.6	9.8	10.0	10.6	14.1	14.7	13.0
		MCWBR	7.9	8.4	9.0	8.3	7.2	5.4	3.9	4.0	4.8	7.4	9.4	9.2
	5% WB	MCDBR	8.8	10.7	12.2	12.7	11.4	8.4	8.3	8.4	8.4	11.6	11.8	12.2
MCWBR		7.6	8.1	8.6	7.8	6.5	4.7	3.9	4.1	4.6	6.9	8.6	9.3	
Clear-Sky Solar Irradiance	taub	0.456	0.496	0.569	0.586	0.603	0.666	0.621	0.569	0.506	0.479	0.465	0.455	
	taud	1.917	1.855	1.713	1.723	1.735	1.679	1.830	1.956	2.064	2.002	1.970	1.929	
	Ebn at Noon	698	722	706	723	718	673	701	729	756	731	685	666	
	Edh at Noon	149	178	223	231	231	244	209	181	155	151	141	140	
All-Sky Solar Radiation	RadAvg	2.43	3.22	4.36	5.05	5.51	4.91	4.53	4.66	4.17	3.70	2.61	2.15	
	RadStd	0.19	0.29	0.39	0.40	0.59	0.45	0.62	0.64	0.41	0.32	0.28	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A

Nomenclature: See separate page